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Studierichting: Informatica
Oefeningenreeks 6G nr. 2.5

$f(x) = x^3 - x^2$

$f(-x) = (-x)^3 - (-x)^2 = -x^3 - x^2$

$f(-x) \neq f(x)$

Dus f is niet even.

$-f(x) = -(x^3 - x^2) = -x^3 + x^2$

$f(-x) \neq -f(x)$

Dus f is ook niet oneven.

We ontbinden:

$$f_E(x) = \frac{f(x) + f(-x)}{2} = -x^2,$$
$$f_O(x) = \frac{f(x) - f(-x)}{2} = x^3$$

a_0 berekenen:

$$a_0 = \frac{2}{\pi} \int_0^\pi (-x^2) dx = -\frac{2}{\pi} \left[\frac{x^3}{3} \right]_0^\pi = -\frac{2\pi^2}{3}$$

a_n berekenen:

$$a_n = \frac{2}{\pi} \int_0^\pi (-x^2) \cos(nx) dx$$

Partiële integratie:

$$\begin{cases} u = x^2 \\ dv = \cos(nx)dx \end{cases} \Rightarrow \begin{cases} du = 2xdx \\ v = \dfrac{\sin(nx)}{n} \end{cases} :$$
$$= -\frac{2}{\pi} \left(\left[\frac{x^2 \sin(nx)}{n} \right]_0^\pi - \frac{2}{n} \int_0^\pi x \sin(nx) dx \right) = \frac{4}{n\pi} \int_0^\pi x \sin(nx) dx$$

Partiële integratie:

$$\begin{cases} u = x \\ dv = \sin(nx)dx \end{cases} \Rightarrow \begin{cases} du = 1dx \\ v = -\dfrac{\cos(nx)}{n} \end{cases} :$$
$$\int_0^\pi x \sin(nx) dx = \left[-\frac{x \cos(nx)}{n} \right]_0^\pi + \frac{1}{n} \left[\frac{\sin(nx)}{n} \right]_0^\pi = \frac{(-1)^{n+1}\pi}{n}$$

Dus:

$$a_n = \frac{4}{n\pi} \cdot \frac{(-1)^{n+1}\pi}{n} = \frac{4(-1)^{n+1}}{n^2}$$

b_n berekenen:

$$b_n = \frac{2}{\pi} \int_0^\pi x^3 \sin(nx) dx$$

Partiële integratie:

$$\begin{cases} u = x^3 \\ dv = \sin(nx)dx \end{cases} \Rightarrow \begin{cases} du = 3x^2 dx \\ v = -\dfrac{\cos(nx)}{n} \end{cases} :$$

$$= \frac{2}{\pi}\left(\left[-\frac{x^3\cos(nx)}{n}\right]_0^\pi + \frac{3}{n}\int_0^\pi x^2 \cos(nx)dx\right)$$

$$= \frac{2}{\pi}\left(\frac{(-1)^{n+1}\pi^3}{n} + \frac{3}{n}\cdot\frac{2(-1)^n\pi}{n^2}\right)$$

$$= \frac{2(-1)^{n+1}\pi^2}{n} + \frac{12(-1)^n}{n^3} = \frac{-2(-1)^n\pi^2}{n} + \frac{12(-1)^n}{n^3} = \frac{2(-1)^n(6-n^2\pi^2)}{n^3}$$

Fourierreeks:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\big(a_n\cos(nx) + b_n\sin(nx)\big)$$

$$f(x) = -\frac{\pi^2}{3} + \sum_{n=1}^{\infty}\left(\frac{4(-1)^{n+1}}{n^2}\cos(nx) + \frac{2(-1)^n(6-n^2\pi^2)}{n^3}\sin(nx)\right)$$

References